

High-Dimensional Probability and Applications in Data Science

Personal Notes

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1 The Curse of Dimensionality and the Monte Carlo Method

High-dimensional probability studies random objects with many coordinates. The key distinction is between many observations and many coordinates per observation. A method that works well with more data points may become too expensive when each point has many coordinates. Random sampling can help.

1.1 Big Data and High Dimension

A data set can be large in two independent ways. It may contain many data points, or each data point may itself have many coordinates. A list of incomes for a large population has many observations but only one coordinate per observation. In contrast, a 100×100 grayscale image is naturally a vector with 10^4 coordinates, one for each pixel. Text, sound, video, genomic data, medical histories, and game records give further examples of high-dimensional observations.

The number of observations often appears linearly in an algorithm or estimator. Dimension is more dangerous: a procedure that resolves every coordinate separately may require a number of operations exponential in the dimension.

Definition 1.1 [*Curse of dimensionality*]. The *curse of dimensionality* means that the work or data needed for a problem can grow exponentially with the number of coordinates. This paraphrases the grid example in the source (p. 2).

1.2 Deterministic Integration on a Grid

Consider numerically approximating the integral of a function $f : [0, 1]^d \rightarrow \mathbb{R}$:

$$I(f) = \int_{[0,1]^d} f(x) dx.$$

In one dimension, a Riemann sum evaluates f on a grid. If the spacing between adjacent points is ε , then the interval needs approximately $1/\varepsilon$ points. Applying the same idea independently to d coordinates produces a Cartesian grid with

$$N \approx \left(\frac{1}{\varepsilon}\right)^d$$

points. Even a moderate resolution becomes infeasible when d is large. For example, ten points per coordinate require 10^d function evaluations.

The same phenomenon is visible geometrically. The unit cube $[0, 1]^d$ has volume 1, while doubling

every side produces a cube of volume

$$\text{vol}([0, 2]^d) = 2^d.$$

There is exponentially more volume to cover as dimension grows. A regular grid pays for this expansion by placing points in every combination of coordinate locations.

Note. A grid includes every combination of coordinate values. With ten choices in each of d coordinates, it needs 10^d points. That multiplication is why the grid grows so quickly.

1.3 Random Sampling

Probability offers a different approximation. Rather than cover the cube with a regular grid, draw independent random points

$$X_1, \dots, X_n \sim \text{Unif}([0, 1]^d)$$

and average the observed function values.

Definition 1.2 [*Monte Carlo estimator*]. The Monte Carlo estimator of $I(f)$ based on n independent uniform samples is

$$\widehat{I}_n(f) = \frac{1}{n} \sum_{i=1}^n f(X_i).$$

Three probability facts explain why this works. If a random vector X has density p on $\mathbb{R}^d$, then

$$\mathbb{E}f(X) = \int_{\mathbb{R}^d} f(x)p(x) dx.$$

Expectation is linear, so $\mathbb{E}(X_1 + \dots + X_n) = \mathbb{E}X_1 + \dots + \mathbb{E}X_n$ without any independence assumption. Variances add when the random variables are independent:

$$\text{Var}(X_1 + \dots + X_n) = \sum_{i=1}^n \text{Var}(X_i).$$

Finally, the strong law of large numbers states that if $Y_1, Y_2, \dots$ are independent and identically distributed with $\mathbb{E}|Y_1| < \infty$, then

$$\frac{1}{n} \sum_{i=1}^n Y_i \longrightarrow \mathbb{E}Y_1 \quad \text{almost surely.}$$

Almost-sure convergence means convergence “with probability = 1” (source, p. 3): the exceptional outcomes have probability zero.

1.4 Unbiasedness and Convergence

The density of a uniform random vector X on $[0, 1]^d$ is 1 inside the cube and 0 outside. Consequently,

$$\mathbb{E}f(X) = \int_{[0,1]^d} f(x) dx = I(f).$$

Linearity of expectation then gives

$$\mathbb{E}\widehat{I}_n(f) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}f(X_i) = I(f).$$

Thus the Monte Carlo estimate is *unbiased*: its expected value is the true integral. If $f(X)$ is integrable, the strong law also gives

$$\widehat{I}_n(f) \longrightarrow I(f) \quad \text{almost surely.}$$

The variance tells us how much the estimate typically differs from the integral.

Proposition 1.3 [*Mean-square error of Monte Carlo integration*]. Suppose $X_1, \dots, X_n$ are independent and uniform on $[0, 1]^d$, and $\text{Var}(f(X_1)) < \infty$. Then

$$\mathbb{E} \left[\left(\widehat{I}_n(f) - I(f) \right)^2 \right] = \frac{\text{Var}(f(X_1))}{n}.$$

Proof. Unbiasedness implies that the mean-square error equals the variance of the estimator.

Independence and identical distribution yield

$$\begin{aligned}\mathbb{E}\left[\left(\widehat{I}_n(f) - I(f)\right)^2\right] &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^n f(X_i)\right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(f(X_i)) \\ &= \frac{\text{Var}(f(X_1))}{n}.\end{aligned}$$

□

If $|f(x)| \leq 1$ throughout the cube, then $\text{Var}(f(X_1)) \leq \mathbb{E}f(X_1)^2 \leq 1$. The root-mean-square error therefore satisfies

$$\left(\mathbb{E}\left[\left(\widehat{I}_n(f) - I(f)\right)^2\right]\right)^{1/2} \leq \frac{1}{\sqrt{n}}.$$

The bound contains no explicit dependence on d . Achieving root-mean-square error at most ε requires $n \geq \varepsilon^{-2}$ samples, compared with roughly ε^{-d} grid points for coordinate-wise resolution ε . For $|f| \leq 1$, the sampling-error bound is independent of dimension. The grid comparison concerns spacing, not the same error guarantee; the cost of sampling a point and evaluating f may still depend on d .

Note. Monte Carlo points do not attempt to cover the cube. They turn integration into estimation of an expectation, so independence controls error through variance rather than geometric coverage.

Source: Roman Vershynin, *High-Dimensional Probability and Applications in Data Science*, Lecture 1, “The Curse of Dimensionality and the Monte Carlo Method,” official course page

<https://www.math.uci.edu/~rvershyn/teaching/hdp/hdp.html> and lecture notes

<https://www.math.uci.edu/~rvershyn/teaching/hdp/Lecture1.pdf>.